
Open Source Web Development

(MSC – ICT Sem - 3)

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Subject : (304) Open Source Web Development

Assignment : Practical Assignment -1

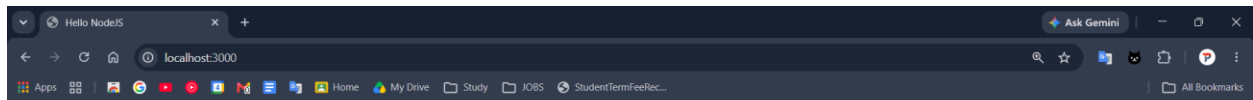
Roll No : 27

Semester : 3rd

Faculty : Ms. Rupal Panchal

1. Develop nodejs application with following requirements:

- Develop a route `"/gethello"` with GET method. It displays **"Hello NodeJS!!"** as response.
- Make an HTML page and display.
- Call `"/gethello"` route from HTML page using AJAX call. (Any frontend AJAX call API can be used.)



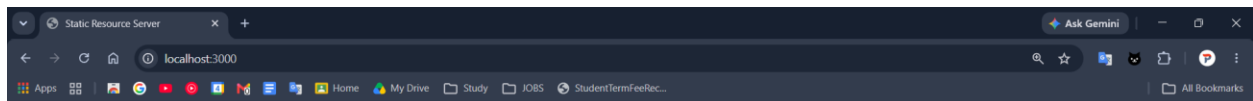
NodeJS AJAX Example

Get Message

Hello NodeJS!!

2. Develop a web server which serves static resources.

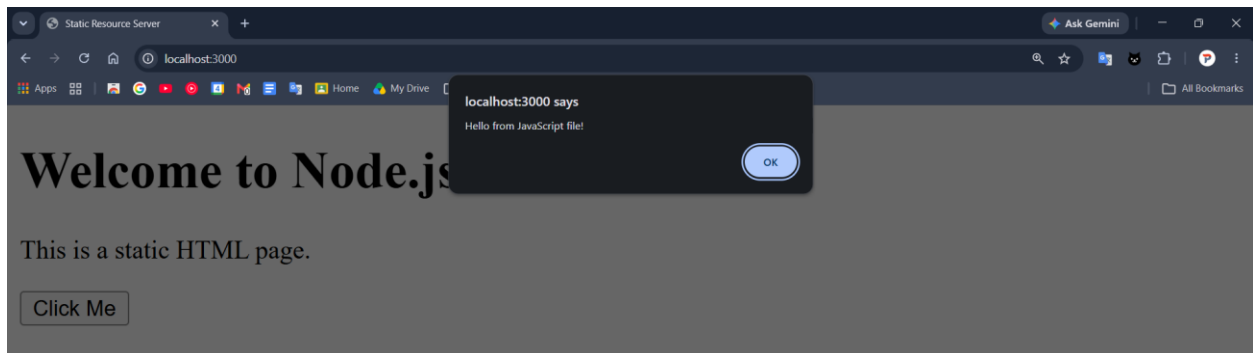
Output :



Welcome to Node.js Server

This is a static HTML page.

Click Me



3. Develop a module for domain specific chatbot and use it in a command line application.

Output:

```
PROBLEMS  OUTPUT  TERMINAL  PORTS  DEBUG CONSOLE
powershell - 3_chatbot + v

PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1> cd 3_chatbot
PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\3_chatbot> node app.js
=== College Chatbot ===
Ask a question (course, fees, admission, contact): course

Bot: We offer BCA, BSc, and BCom.

PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\3_chatbot> node app.js
=== College Chatbot ===
Ask a question (course, fees, admission, contact): contact

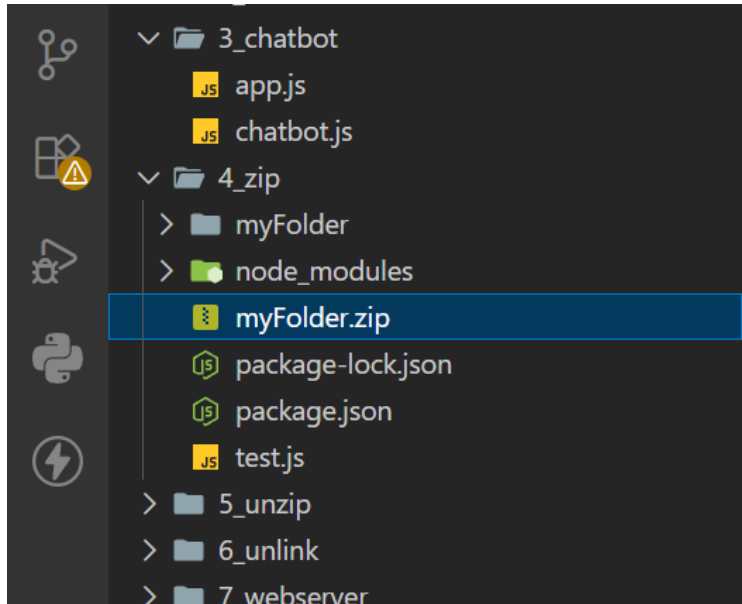
Bot: Call us at 9876543210.

PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\3_chatbot> |
```

4. Write a program to create a compressed zip file for a folder.

Output

```
PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1> cd 4_zip
PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\4_zip> node test.js
ZIP file created successfully!
PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\4_zip> |
```



```
3_chatbot
├── app.js
├── chatbot.js
└── 4_zip
    ├── myFolder
    ├── node_modules
    ├── myFolder.zip
    ├── package-lock.json
    ├── package.json
    └── test.js
5_unzip
6_unlink
7_webserver
```

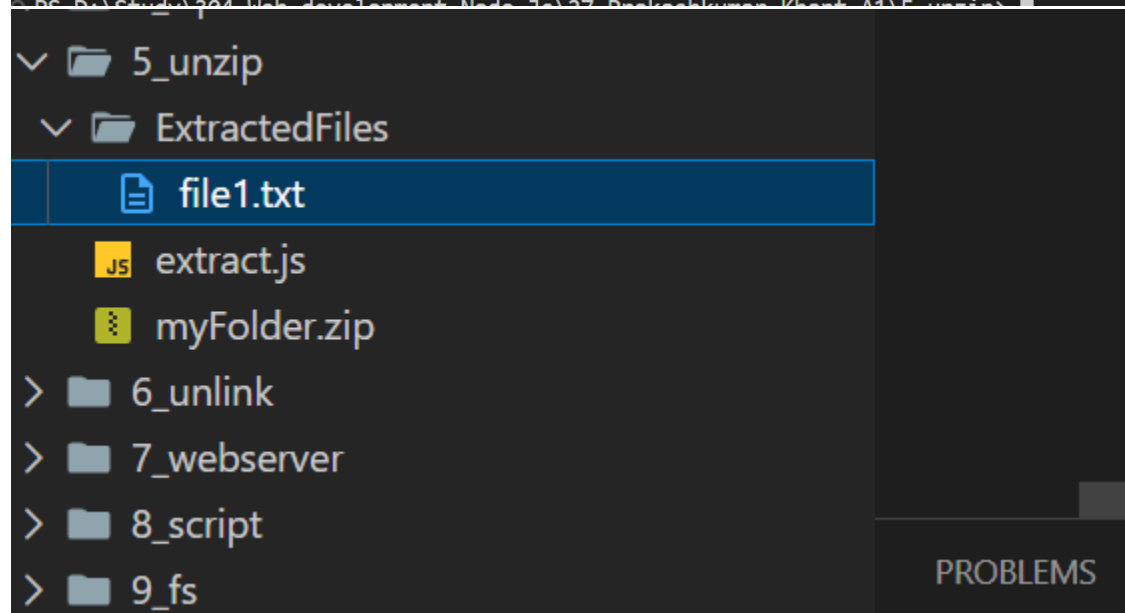
PROBLEMS OUTPUT TERMINAL P

PS D:\Study\304 Web development

5. Write a program to extract a zip file.

Output

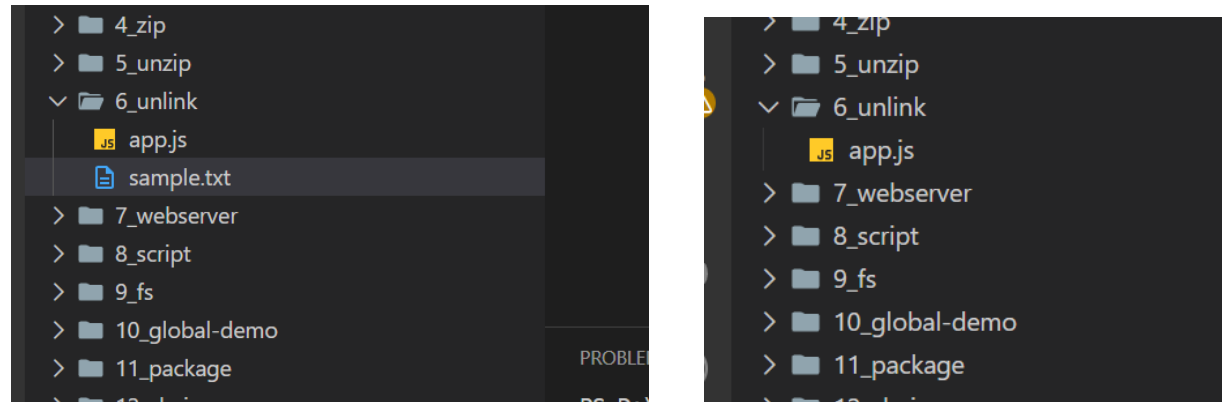
```
PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\5_unzip> node extract.js
ZIP file extracted successfully!
```



The screenshot shows the VS Code Explorer with the '5_unzip' folder expanded. Inside, there is a subfolder named 'ExtractedFiles' which contains a file named 'file1.txt'. Other files visible in the same directory are 'extract.js' and 'myFolder.zip'. The Explorer also shows other folders like '6_unlink', '7_webserver', '8_script', and '9_fs'.

6. Write a program to promisify fs.unlink function and call it.

Output



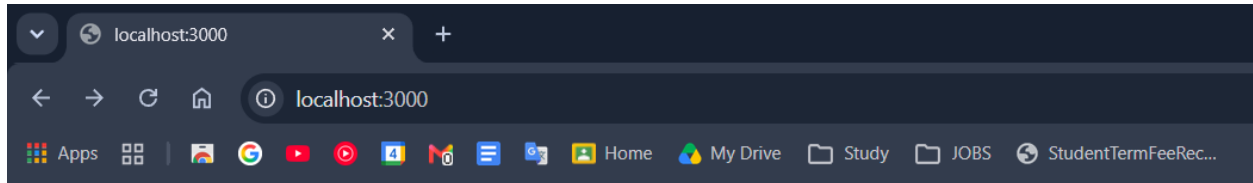
```
\sample.txt'
```

```
PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\6_unlink> node unlink.js
File deleted successfully
```

The output shows two side-by-side screenshots of the VS Code Explorer. The left screenshot shows the file structure before deletion, with 'sample.txt' highlighted in the '6_unlink' folder. The right screenshot shows the same structure after deletion, with 'sample.txt' missing. Below the screenshots, a terminal window shows the command 'node unlink.js' being executed, resulting in the output 'File deleted successfully'.

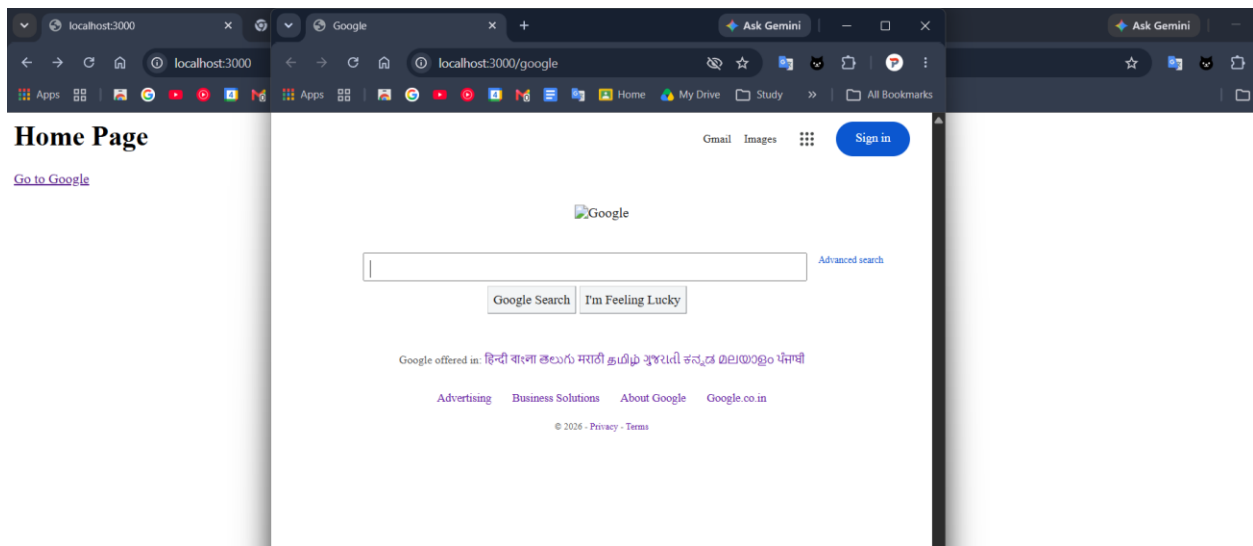
7. Develop a node web server with a route /google which can fetch data of google page using fetch() using async-await model and show as a response.

Output



Home Page

[Go to Google](#)



8. Set a server script, a test script and 3 user defined scripts in package.json file in your nodejs application.

Output

```
PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\8_script> cd my-node-app
PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\8_script\my-node-app> node server.js
Server running at http://localhost:3000
```

test.jsapp.jspackage.json X

8_script > my-node-app > package.json > ...

```
1  {
2    "name": "my-node-app",
3    "version": "1.0.0",
4    "description": "Node.js Scripts Example",
5    "main": "index.js",
6    "scripts": {
7      "server": "node server.js",
8      "test": "echo \"Test Successful\"",
9      "start": "node index.js",
10     "dev": "nodemon server.js",
11     "build": "echo \"Build Completed Successfully\""
12   },
13   "author": "",
14   "license": "ISC",
15   "devDependencies": {
16     "nodemon": "^3.0.0"
17   }
18 }
```

● PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\8_script\my-node-app> npm test

> my-node-app@1.0.0 test

> echo "Test Successful"

"Test Successful"

○ PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\8_script\my-node-app>

○ PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\8_script\my-node-app> npm run dev

> my-node-app@1.0.0 dev

> nodemon server.js

[nodemon] 3.1.14

[nodemon] to restart at any time, enter `rs`

[nodemon] watching path(s): *.*

[nodemon] watching extensions: js,mjs,cjs,json

[nodemon] starting `node server.js`

Server running at http://localhost:3000

● PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\8_script\my-node-app> npm run build

> my-node-app@1.0.0 build

> echo "Build Completed Successfully"

"Build Completed Successfully"

○ PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\8_script\my-node-app>

9. A program which calls useful functions in fs module.

Output

```
● PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1> cd 9_fs
● PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\9_fs> node app.js
File created successfully.
File Content: Hello, this is a sample file.
Data appended successfully.
File renamed successfully.
File exists.
File Size: 52 bytes
File deleted successfully.
○ PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\9_fs> |
```

10. A program which uses global objects in nodejs. Also read command line arguments and print it.

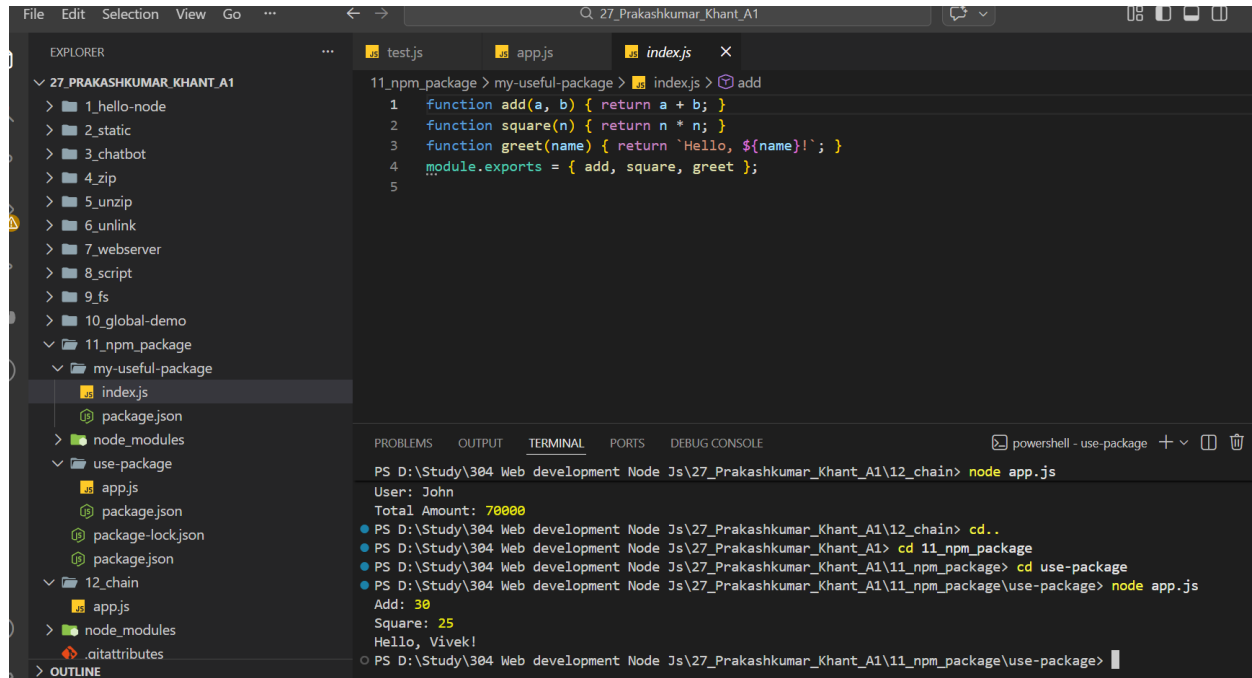
Output

```
● PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\10_global-demo> node app.js
=== Global Objects in Node.js ===
Current File: D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\10_global-demo\app.js
Current Directory: D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\10_global-demo
Operating System: win32
Node.js Version: v24.15.0
Current Working Directory: D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\10_global-demo

=== Command Line Arguments ===
No command line arguments provided.
○ PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\10_global-demo> |
```

11. Develop a useful package and publish it on npmjs.com. Write a program to use it.

Output



The screenshot shows a VS Code editor with a project structure on the left and a terminal at the bottom. The project structure includes a folder named '11_npm_package' containing a 'my-useful-package' subfolder. The 'my-useful-package' folder contains an 'index.js' file and a 'package.json' file. The 'index.js' file contains the following code:

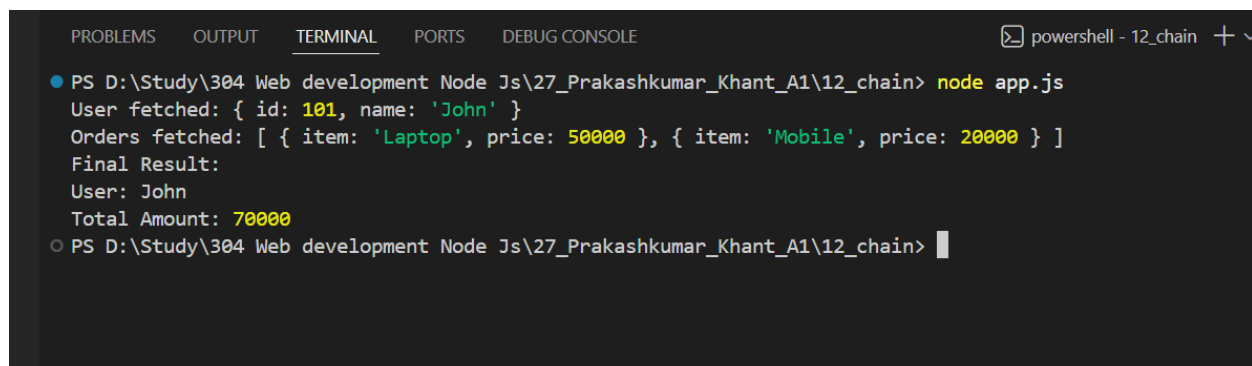
```
1 function add(a, b) { return a + b; }
2 function square(n) { return n * n; }
3 function greet(name) { return `Hello, ${name}!`; }
4 module.exports = { add, square, greet };
5
```

The terminal shows the following commands and output:

```
PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\12_chain> node app.js
User: John
Total Amount: 70000
PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1> cd 11_npm_package
PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\11_npm_package> cd use-package
PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\11_npm_package\use-package> node app.js
Add: 30
Square: 25
Hello, Vivek!
```

12. Write a node program having at least 3 dependent tasks and call using chained promises.

Output



The screenshot shows a VS Code terminal window with the following output:

```
PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\12_chain> node app.js
User fetched: { id: 101, name: 'John' }
Orders fetched: [ { item: 'Laptop', price: 50000 }, { item: 'Mobile', price: 20000 } ]
Final Result:
User: John
Total Amount: 70000
PS D:\Study\304 Web development Node Js\27_Prakashkumar_Khant_A1\12_chain>
```